

Architectural Specification: Reference & Bias Cell Integration

HARDWARE DESIGN BASIS FOR INVERTED MAML ANALOG COMPUTE MATRIX

1. Executive Summary & Core Design Philosophy

In modern Analog Compute-in-Memory (CiM) architectures based on 2T1C (Two Transistors, One Capacitor) topologies, physical non-idealities present a severe bottleneck to calculation precision. Chief among these non-idealities is deterministic charge leakage, commonly referred to as **voltage droop**. Traditional implementations suffer from catastrophic drift because the physical weights degrade asymmetrically over time.

This architecture circumvents this constraint by deploying an **Inverted Meta-Agnostic Meta-Learning (MAML)** algorithm in the digital control layer. Rather than forcing the analog medium to maintain flawless charge preservation, the algorithm actively treats the hardware's temporal degradation as a predictable, compensable phenomenon. This document specifies the implementation of dedicated hardware structures—namely **Reference Cells** and **Hardware Bias Cells**—designed to serve as real-time calibration anchors. By exploiting customized RC discharge trajectories, the system derives a localized, normalized temporal clock directly from the kisel/substrate dynamics, enabling digital precision over an inherently non-deterministic analog medium.

2. Reference Cells Architecture & Accelerated Droop Control

The primary objective of the Reference Cells is to provide a comprehensive, wide-swing representation of the charge degradation curve across the 10 ms refresh cycle. Instead of preserving state, these cells are deliberately engineered to sweep across the majority of the active data window, acting as a dynamic "measuring stick" for the digital error estimation engine.

2.1 Elimination of Input Data-DAC Complexity

To ensure maximum Signal-to-Noise Ratio (SNR) and isolate the weight degradation factor completely, the input data lines (V_{in}) for the Reference Cells are decoupled from the digital-to-analog converter (DAC) matrix lines. The data row input is permanently tied to a fixed reference potential of **1.90 V**, which represents the absolute maximum data magnitude (Data 255).

Architectural Advantage: Hardwiring the input data line to a clean 1.90 V via a dedicated, low-noise operational amplifier (such as the MCP6024T) removes routing complexity, conserves FPGA I/O resources, eliminates DAC non-linearities from the reference baseline, and yields a maximum output current response ($I_{out} \propto V_{ds}$) that is purely dependent on the instantaneous gate voltage.

2.2 Accelerated RC Trajectory and Sweep Mechanics

The Reference Cells share the standard **100 nF Weight Capacitor (C_w)** but utilize an accelerated discharge path. By integrating a dedicated **330 kΩ urladdningsresistor** connected directly to Ground (0V), the cell forces an aggressive but highly predictable voltage sweep.

The active operational gate voltage range for the synaptic transistor (e.g., DOX3134A) spans from 2.1 V (Data 0) to 3.1 V (Data 255), representing a 1.0 V dynamic range above the virtual ground source potential ($V_s = 1.65\text{ V}$).

Using the fundamental RC discharge equation, we map the temporal behavior of the Reference Cell across the 10 ms refresh window:

$$\tau_{ref} = R \cdot C = 330 \text{ k}\Omega \times 100 \text{ nF} = 33.0 \text{ ms}$$

Initializing the cell at a standard high-drift operational maximum of **3.0 V** (corresponding approximately to Data 230), the remaining voltage at the expiration of the 10 ms cycle is calculated as:

$$V(10 \text{ ms}) = 3.0 \text{ V} \cdot e^{-\frac{10 \text{ ms}}{33.0 \text{ ms}}} = 3.0 \text{ V} \cdot e^{-0.30303} \approx 3.0 \text{ V} \cdot 0.73857 = \mathbf{2.216 \text{ V}}$$

2.3 Dynamic Data Window Mapping

Translating the final voltage of 2.216 V back into the normalized 8-bit digital space (where 2.1 V = Data 0 and 3.1 V = Data 255):

$$\text{Data Value} = \frac{2.216 \text{ V} - 2.1 \text{ V}}{1.0 \text{ V}} \times 255 = 0.116 \times 255 \approx \mathbf{29.6}$$

Thus, the Reference Cell sweeps elegantly from **Data 230 down to Data 30** over the course of the 10 ms refresh interval. This results in an intentional voltage fall of **784 mV**, successfully sweeping through approximately **78.4% of the complete dynamic data swing**. This maximizes the derivative resolution available to the Inverted MAML algorithm while maintaining a safe margin from the hard boundaries (0 and 255) to prevent operational clipping or saturation.

3. Hardware Bias Cells & Synchronous Drift Matching

Conventional artificial neural networks establish neuron stability via a mathematical bias term: $y = \sum (W \cdot x) + b$. To implement this natively within an analog Compute-in-Memory environment without resorting to costly external digital summation or specialized analog summing amplifiers, this architecture embeds dedicated **Hardware Bias Cells** directly into the matrix arrays.

3.1 Input Configuration and Learning Mechanics

Similar to the Reference Cells, the Hardware Bias Cells require a constant input factor to isolate the weight parameter. Consequently, the input data line (V_{in}) for all bias columns/rows is hardwired directly to the **1.90 V reference rail** (Data 255), emulating a permanent input activation magnitude of exactly 1.0.

Crucially, the weight stored on the 100 nF capacitor of the Bias Cell is fully programmable and modifiable by the Inverted MAML control logic. The nätverk is capable of training and adjusting these bias weights during

the standard update phase, granting the network structural degrees of freedom to center its activation functions.

3.2 Baseline Droop Tracking

Unlike the reference cells, the Bias Cells are designed to experience the exact same decay characteristics as the standard payload data-carrying cells. They are configured with the standard system baseline **1 MΩ discharge resistor** connected to GND.

$$\tau_{bias} = R \cdot C = 1 \text{ ext{ M}}\Omega \times 100 \text{ ext{ nF}} = 100.0 \text{ ext{ ms}}$$

Over the 10 ms refresh period, the Bias Cell experiences a highly controlled, standardized charge decay:

$$V(10 \text{ ext{ ms}}) = V_0 \cdot e^{\{- \frac{10 \text{ ext{ ms}}}{100 \text{ ext{ ms}}}\}} = V_0 \cdot e^{\{-0.1\}} \text{ pprox } V_0 \cdot 0.9048$$

This corresponds to a target voltage drop (droop) of exactly ~9.52% relative to the active charge above the ground potential. Because the Bias Cells decay synchronously with the payload cells, any common-mode drift or ambient thermal fluctuations affecting the array are naturally mirrored in the bias terms, allowing the error-correction loop to stabilize the network seamlessly.

4. Matrix Topologies & System Symmetrization

The physical arrangement of these arrays is optimized to maintain geometric symmetry, simple FPGA routing, and localized thermal matching.

4.1 Matrix(3) Topology: Input Adjustment Stage (8 × 6 Array)

Matrix(3) serves as the primary ingress gating matrix. It must process the entire incoming data vector alongside the calibration channels to prevent information truncation. It is structured as an **8 × 6 array** (8 Rows, 6 Columns):

- **Rows 1–6:** Dedicated to primary input payload data streams (variable via input DACs).
- **Row 7 (Reference Row):** Contains the wide-swing Reference Klockor (Hardwired $V_{in} = 1.90 \text{ ext{ V}}$, 330 kΩ leakage path).
- **Row 8 (Bias Row):** Contains the standard network Bias Nodes (Hardwired $V_{in} = 1.90 \text{ ext{ V}}$, 1 MΩ leakage path).

Matrix(3) Layout	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
Rows 1–6 (Data)	Cell (D)	Cell (D)	Cell (D)	Cell (D)	Cell (D)	Cell (D)
Row 7 (Reference Clock)	REF (330k)	REF (330k)	REF (330k)	REF (330k)	REF (330k)	REF (330k)
Row 8 (Hardware Bias)	BIAS (1M)	BIAS (1M)	BIAS (1M)	BIAS (1M)	BIAS (1M)	BIAS (1M)

4.2 Matrix(8) Topology: Deep Compression Stage (7 × 6 Array)

For the secondary/deep computational block, Matrix(8), a **7 × 6 array** (7 Rows, 6 Columns) is selected to maintain architectural symmetrization and optimize learning capacity during the prototyping phase.

- **Rows 1–6:** Core internal network data representations.
- **Row 7 (Bias Row):** Integrated Hardware Bias cells on every column line (Hardwired $V_{in} = 1.90\text{ ext{V}}$, 1 MΩ leakage path).

By opting for a 7 × 6 symmetric architecture over a highly compressed configuration (e.g., 5 × 6 or 4 × 6), the system retains a higher degree of computational freedom. This over-provisioning ensures that the Inverted MAML framework possesses sufficient learning capability to absorb severe spatial hardware mismatches and cross-talk during validation. Space optimization and network pruning can subsequently be executed in future ASIC shrinks.

5. Component and Parameter Summary

Cell Classification	Capacitor (C_W)	Resistor (R_{leak})	Input Voltage (V_{in})	10ms Target Swing	Primary System Function
Payload Data Cell	100 nF	1 MΩ	Variable (DAC) 1.65 V – 1.90 V	~9.52% Drop (~0.1 V relative)	Executes core neural multiplication matrix operations ($W \times x$).
Reference Cell	100 nF	330 kΩ	Fixed (OP-Amp) 1.90 V (Data 255)	~78.4% Wide Sweep (Data 230 → Data 30)	Provides high-resolution temporal decay tracking for the MAML kloka.
Hardware Bias Cell	100 nF	1 MΩ	Fixed (OP-Amp) 1.90 V (Data 255)	~9.52% Drop (Synchronous match)	Injects trained offset parameters ($+b$) with auto-tracking drift compensation.

6. Roadmap Towards Monolithic ASIC Co-Design

While these parameters reflect the physical reality of the discrete PCB prototype (utilizing 100 nF discrete capacitors and 330 kΩ/1 MΩ discharge networks), the underlying architectural principles map directly to sub-micron monolithic kisel fabrication nodes.

In an advanced analog/mixed-signal ASIC implementation, the sweet spot is achieved using **40 nm or 28 nm Planar CMOS** technology rather than fin-based architectures (FinFETs), which introduce discrete quantization steps and severe gate-oxide leakage. On kisel, the massive 100 nF discrete capacitances scale down proportionally to integrated metal-oxide-metal (MOM) or metal-insulator-metal (MIM) structures of approximately **10 fF to 30 fF**.

Correspondingly, the physical discharge resistors are replaced by controlled sub-threshold channel leakage inside the access transistors (measured in pico-amperes or femto-amperes). To accommodate the smaller charge storage capacity, the total refresh window shifts from 10 ms down to a high-speed interval of **10 μ s to 100 μ s**. Because the MAML framework scales purely based on the relative ratio of the time constants, the 78.4% reference sweep and the 9.5% bias matching remain mathematically identical, proving the long-term scalability of this hybrid analog-digital architecture.